PRT564 Data Analytics and Visualisation

Assessment 01:

Outlining a Predictive Modeling and Demand Classification Framework for Multi-Modal Opal Trip Data

Campus Location: Sydney
Submission Date: 20 March 2026

Dataset Link: https://data.gov.au/data/dataset/nsw-2-opal-trips-all-modes/resource/77718b3e-ea01-4401-a3bf-0686edfdb1f2

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1. Introduction

The State organization, Transport for NSW (TfNSW), organises the transport services within New South Wales, Australia. They aim for a sustainable and seamless transport network, for easy travel, work and live along with safety. It manages and plans various types of transport system such as trains, metro systems, buses, ferries, light rail and taxis/ ride share, They focus on the needs of the users. They also work on transport planning, point to point (NSW Government, 2026), creating places and strategies.

1.1 Research Problem

Public are facing problems due to overcrowding, congestion and longer travel times. Moreover, it is expected to have more residents in Sydney, especially in western suburbs, which may cause more uncomfortable and disturbed journey. Users behavior have shown different travel patterns , at some point trains are crowed and at another point buses are crowded. The travel demand is changing according peak and off-peak hours. These frequent pattern change has challenged the organization to plan services properly, manage overcrowding and predict busy periods.

The detailed analysis of passenger travel patterns in different transport modes and time periods can improve planning, reduce overcrowding and make proper strategy for seamless transport services.

Why Data Analysis Helps?

The detailed analysis of data provides insights for travel trends and patterns, seasonality and changes in the transport modes over time periods. Investigation on trips numbers over time provides details to determine the time when the demand of different transport services is high , and decide extra services. Moreover, tap-on and tap-off records will be combined and analysed, while transforming monthly records into daily trip statistics.

Stakeholders:

- **Transport Planners and Policymakers:** Uses the insights for optimising service plans and resource allocation.
- **Passengers and Commuters:** Benefits from improved service reliability, reduced congestion, and better travel experience.
- **Government and Policy makers:** Uses the findings to support infrastructure investment decisions and long-term transport policies.

2. Dataset Description and Suitability Assessment

2.1 Dataset Overview

Australian Government published the dataset "Opal Trip Counts Bases on NRT – by month" on the website data.gov.au (Government, 2025). It represents aggregated monthly public transport use in New South Wales (NSW) between July 2026, including trips made on the use of Opal cards on buses, trains, ferries, light rail, and metro. The dataset contains 6656 records with four key variables, available in a CSV file. It can be extracted, structured, and analysed accordingly.

2.2 Dataset Structure

The dataset consists of four key variables that describe various features of the transport use.

- **Year_Month** (temporal, string): period of observation; 116 unique months.
- **Card_Type** (categorical, string): passenger categories (Adult, Concession, Student, etc.); 26 unique values including UNKNOWN.
- **Travel_Mode** (categorical, string): transport mode (Bus, train, Ferry, Light Rail, Metro); 11 unique values including UNALLOCATED.
- **Trip_Count** (numerical, float): monthly trip counts; min=0, max (approx.) 27.9 million.

The dataset has no missing values, however, the number of rows with unknown card types and unallocated travel modes are 115 and 38, representing unclassified or system-documented trips. There weren't any duplicates of data. The inconsistencies in categorical values in travel mode (i.e "Train" vs "train") is present, affecting accuracy and efficiency. The data standardisation will overcome this problem for seamless comparison across categories.

Dataset Preview (First 8 Records)

Year_Month	Card_type	Travel_Mode	Trip
Jul-2016	Adult	Bus	13146432.0
Jul-2016	Child/Youth	Bus	1079640.0
Jul-2016	Concession	Bus	1845322.0
Jul-2016	Employee	Bus	64989.0
Jul-2016	Free Travel	Bus	25228.0
Jul-2016	School Student	Bus	1297240.0
Jul-2016	Senior/Pensioner	Bus	3739658.0
Jul-2016	Adult	Ferry	861272.0

Figure 1 - Sample preview of the Opal dataset showing key variables and records

2.3 Suitability for Analysis

The dataset contains temporal, categorical data, and numerical variables, which allows tread analysis and comparison across transport modes while evaluating the travel patterns over time. It contains data from the year 2016 to 2025, considering a significant coverage, will give insights about seasonal trends and demand variations.

For the detailed analysis, the dataset will be combined with population statistics provided by Australian Bureau of Statistics (ABS), allowing data analysis per captia. The dataset will be matched with time and geographical territory.

Although, Missing proper data entries (i.e "UNKNOWN" and "UNALLOCATED"), the absence of spatial features, and the data feature monthly aggregation , not allowing detailed analysis of peak times, the dataset provides detailed analysis and building predictive models.

3. Research questions and data analytic objectives

3.1 Research Questions

1) Has the total number of Opal trips changed over time across different transport modes?
2) Which public transport modes have the highest passenger usage?
3) How is the seasonal or monthly trends in demands of public transport ?
4) How do different card types (e.g., adult, concession) influence travel patterns?
5) Which transport modes show increasing or declining usage trends?

3.2 Type of Analysis

Descriptive analysis is used to summarise statistics and explore trends and patterns in public transport usage, along with elements of predictive analysis to forecast future demand.

3.3 Organisational Benefit

Transport for NSW will have insights for passenger travel patterns, which is a step to optimise transport services. It will lead improved service planning, resource allocation, and policy decisions to enhance public transport efficiency with commuter satisfaction.

4. Data Analysis Methods

The project will utilize the combined dataset to analyse the total number of Opal trips changed over time across different transport modes, public transport with highest usage, seasonal and monthly trends, card type and their travel patterns, and trends. This analysis combines a scripted ETL pipeline with exploratory data visualization, descriptive statistics, regression, and classification models chosen to match the data structure.

4.1 ETL plan and automated extraction

- **Method:** ETL (Python/ pandas)
- **Justification:** Standardisation of variables along with data cleaning will ensure accurate comparison for travel modes across time. Combination of population data with opal serves deeper insights for transport usage.
- **Extract:** Download "*Opal Trip Counts Bases on NRT – by month*" dataset from data.gov.au; and "*Population – states and territories (Quarterly ERP)"* from the Australian Bureau of Statistics (Australian Bureau of Statistics, 2026).
- **Transform:** Standardisation of date formats and categorical variables (transport mode, card type), Aggregate trip data (monthly totals), Feature engineering (Peak/off- peak indicators).
- **Load:** analysis-ready CSV dataset

4.2 Regression

- **Method:** Analyse the relationship between dependent variable (i.e Trip count (demand)), and independent variables (i.e Time, transport mode, and card type)
- **Models:** Linear regression, Poisson regression (for count data like trips)
- **Justification:**
    - Transport demands refers to count data, enabling Poisson models. Poisson regression can give insights on factors that can predict the frequency of an event (ScienceDirect, n.d.).
    - Supports trend estimation and prediction.

- **Assumptions:** The target variable follows a count distribution. Observations are independent, where predictors are not highly correlated.
- **Verification:** Mean-variance check, residual diagnostics to evaluate model fit, and auto correlation check.

4.3 Descriptive Statistics

a) **Method:** Aggregation (sum, mean, count), grouping by variables (mode, time, card type)

b) **Justification:** Descriptive analysis for total trips, average usage, and distribution patterns, allowing the comparison across transport modes and time periods. It enables understanding patterns and trends

c) Assumptions:

- There are no/negligible biases in the data.
- Aggregation accurately reflects overall travel behaviour.
- There aren't any significant outliers.

4.4 Data Visualisation

- **Method:** Time-series lines, heat maps, bars, histogram (Matplotlib / seaborn/ pyplot).
- **Justification:** The visualization of peak hours / peak months for public transport usage by Time-series lines. In addition, Heatmaps also be used for peak travel time and the most used train line, while bar illustrates the most used travel mode.
- **Assumptions:** Tap-on and tap-off data are accurate for actual passenger demand.
- **Limitation:** Track work, weekends, and public holidays are not considered.

5. Task and Work Plan

5.1 Roles

Role	Notes
Organisation & Research	organisation details, problem definition, and research questions
Methods & Project Management	data analytics method, project planning, overall project timeline/milestones
Dataset & Data Quality	Dataset description, quality, ethics.

Figure 2 - Team Roles Table

5.2 Task Planning

Task	Responsibility	Notes
Organization Context & Problem Definition	organization description, define business/research problem/explain why data analysis helps, identify stakeholders	understanding organizational context
Dataset Description & Suitability Assessment	Describe dataset, its structure, variable, data types, file format, discuss strength, limitations, bias, ethical/ privacy consideration	Dataset understanding
Research question & analysis objectives	Formulate research questions, type of analysis for each, expected organizational benefit	Problem Statement analysis
Data Analysis Methods	Explain and justify methods including descriptive statistics, regression, classification visualization, assumptions, verifications	Data processing and insights interpretation

Figure 3 - Responsibility Table

5.3 Timeline/milestones

Gantt Chart for Project

Task	Mar 11	Mar 13	Mar 15	Mar 17	Mar 19	Mar 21
Data Selection and Registration						
Organisation Context and problem definition						
Data Description						
Research Questions						
Data Analysis Methods						
Workplan						
Final Review						

Figure 4t- Gantt Chart: Data Analytic Planning

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11
Planning											
Data preprocessing											
EDA											
Descriptive Statistics											
Feature Engineering											
Regression Models											
Evaluation											

Figure 5 - Project Gantt Chart of the overall Project Plan

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